



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: Computer Networks Course Code: CS305

Department : Department of Computer Science and Engineering

Exam Paper Setter: _____

Question No.	1	2	3	4	5	6	7	8	9	10
Score										

Problem I. True or False (T or F, 2 points each)

1. ____ At routers, packets queue at input ports due to the limited transmission rate of the output ports.
2. ____ To achieve reliable data transfer, error detection is needed.
3. ____ Data plane corresponds to per-router behaviors, and control plane corresponds to end-to-end data transfer.
4. ____ From the network perspective, an end system is mobile if it moves from one room to another.
5. ____ Since TCP socket is identified by destination port number, destination IP address, source port number, and source IP address, a TCP segment contains the fields of destination IP address and source IP address.
6. ____ IP service model is a best-effort delivery service.
7. ____ Suppose Host A sends one segment with sequence number 38 and 4 bytes of data over a TCP connection to Host B. In this same segment, the acknowledgment number is necessarily 42.
8. ____ DHCP is a client-server protocol.
9. ____ With OSPF route, each router constructs a complete topological map of the entire autonomous system and locally runs Dijkstra's shortest path algorithm.
10. ____ Bluetooth network (that connect small wireless devices such as keyboards and headsets) is a type of single-hop and infrastructure-based networks.

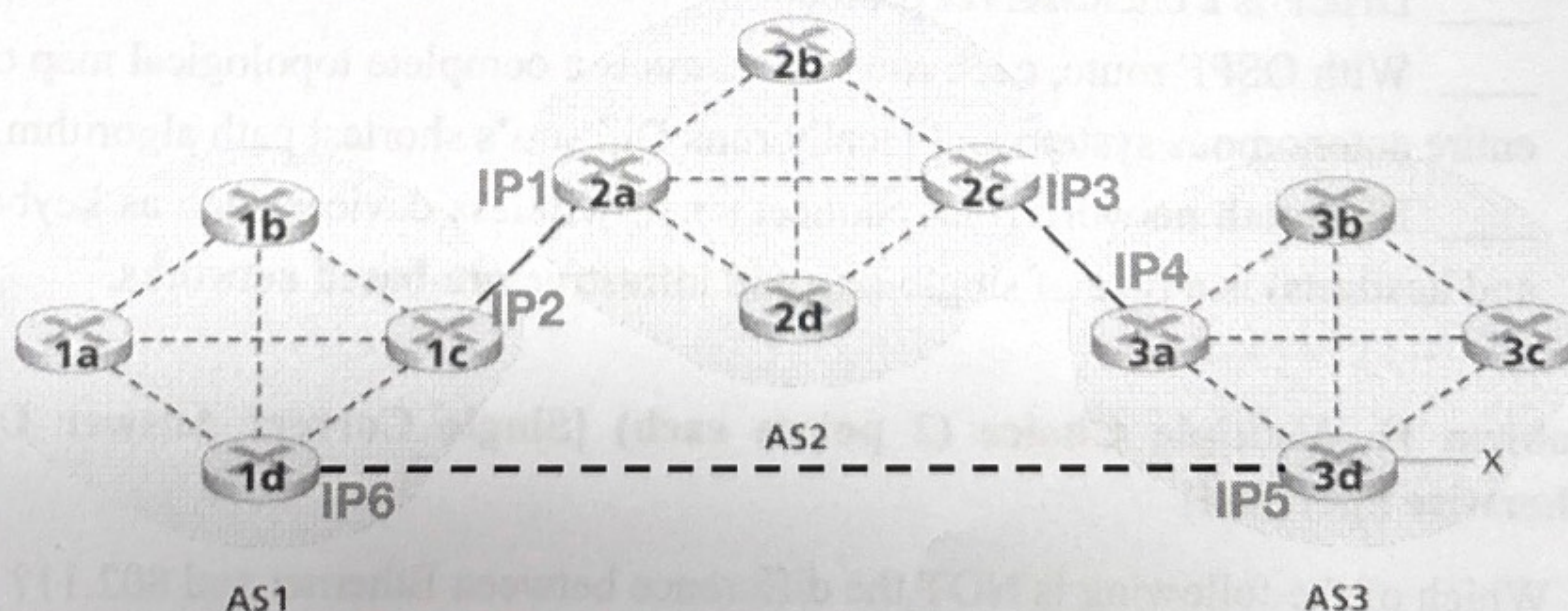
Problem II. Multiple Choice (2 points each) [Single Correct Answer Unless Otherwise Specified]

1. Which of the following is NOT the difference between Ethernet and 802.11?
A. wireless or wired LAN technology B. reliable or unreliable

- C. use CSMA/CD or CSMA/CA D. connection or connectionless
2. Host A initializes a TCP connection with Host B. In the SYN segment sent by host A, the sequence number is 900. To disconnect the connection, host A sends a FIN segment with sequence number 2001. There is no retransmission. The third stage of the three-way handshake carries application-layer data. How many bytes of application-layer data have host A sent to Host B during the connection?
- A. 1100 B. 1099 C. 1101 D. None of above
3. Which of the following is true about Autonomous System (AS) routing?
- A. Each AS may run its own intra-AS and inter-AS routing protocols.
 B. Each AS may run its own intra-AS routing protocol, but they all run the same inter-AS routing protocol (BGP).
 C. All Internet routers run exactly one routing protocol (BGP).
 D. None of the above.
4. Consider the following forwarding table. What are the interfaces for 217.26.128.196, 217.26.128.1, 217.26.128.136 respectively?

Destination Address Range	Link interface
217.26.128.0/24	1
217.26.128.128/25	2
217.26.128.192/26	3
Otherwise	4

- A. 3; 1; 2 B. 3; 1; 4 C. 4; 1; 3 D. None of above
5. Which of the following function(s) can be achieved by generalized forwarding?
[Unknown number of correct answers; mark all correct answers]
- A. Layer-2 switching B. Firewall C. Load-balancing D. NAT
6. Consider the network shown below. Notations "IP1", "IP2", ... correspond to the IP addresses of the corresponding interface. What are the two message received by router 1b in AS1 about the reachability information of x?



- A. [IP4; AS3; x] and [IP5; AS3; x]

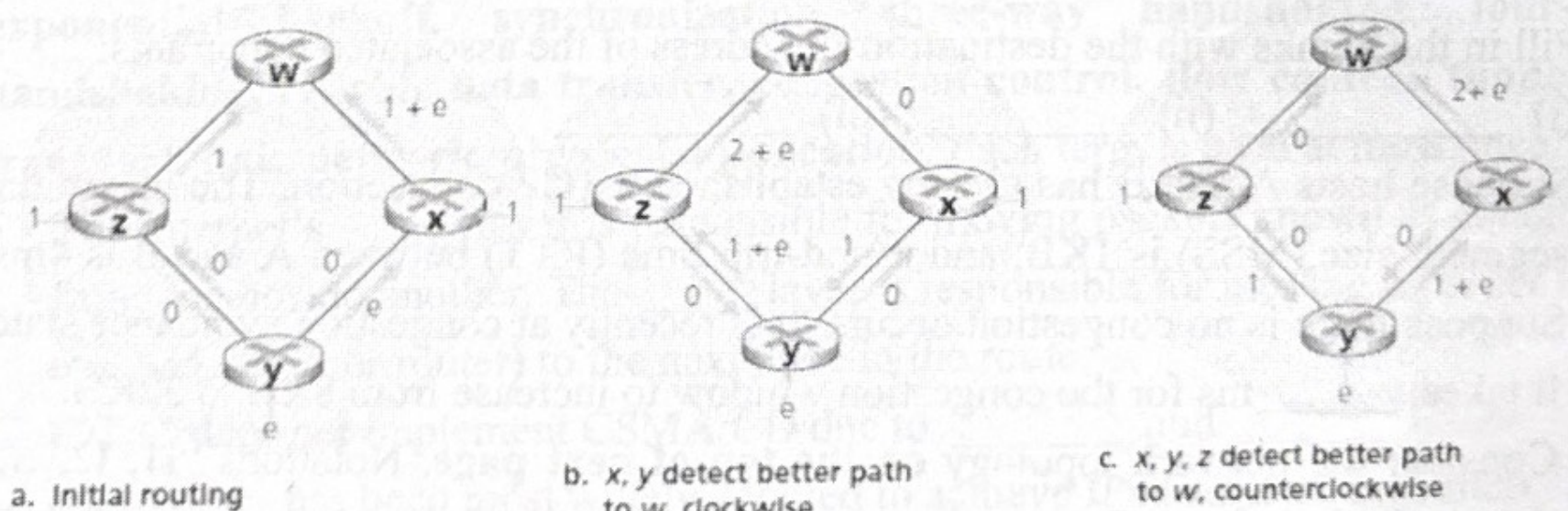
- B. [IP2; AS2, AS3; x] and [IP6; AS3; x]
 C. [IP1; AS2; x] and [IP5; AS3; x]
 D. [IP1; AS2, AS3; x] and [IP5; AS3; x]
7. Consider the figure shown in question II.6. Suppose the physical link between 1d and 3d is removed. AS1 is running RIP for intra-AS routing. AS2 and AS3 are running OSPF for intra-AS routing. Suppose eBGP and iBGP are used for inter-AS routing. Which protocols do routers 2c, 2d, 1c, 1d learn about prefix x from, respectively?
- A. eBGP; OSPF; eBGP; RIP B. iBGP; eBGP; iBGP; eBGP
 C. eBGP; iBGP; eBGP; iBGP D. None of above

8. Consider the above figure in question II.6. Suppose 1b has a forwarding table shown below, and it recently adds an entry related to x. BGP's route-selection algorithm is used, and each dash corresponds to a physical link with the same cost. Suppose the router has the same preference over all routes. What is the value of y?

Destination	Interface
IP1	3
IP5	2
x	y
Otherwise	1

- A. 1 B. 2 C. 3 D. 4

9. Consider the oscillation problem shown below. Nodes z, y, and x send data to node w, where the amount of data they send is shown in the figure. The initial routing is shown in subfigure a. Then, in subfigure b, when x and y detect better paths to w, they changed their direction for data forwarding. Which of following method is the **best** one that can solve (or partially solve) this problem?

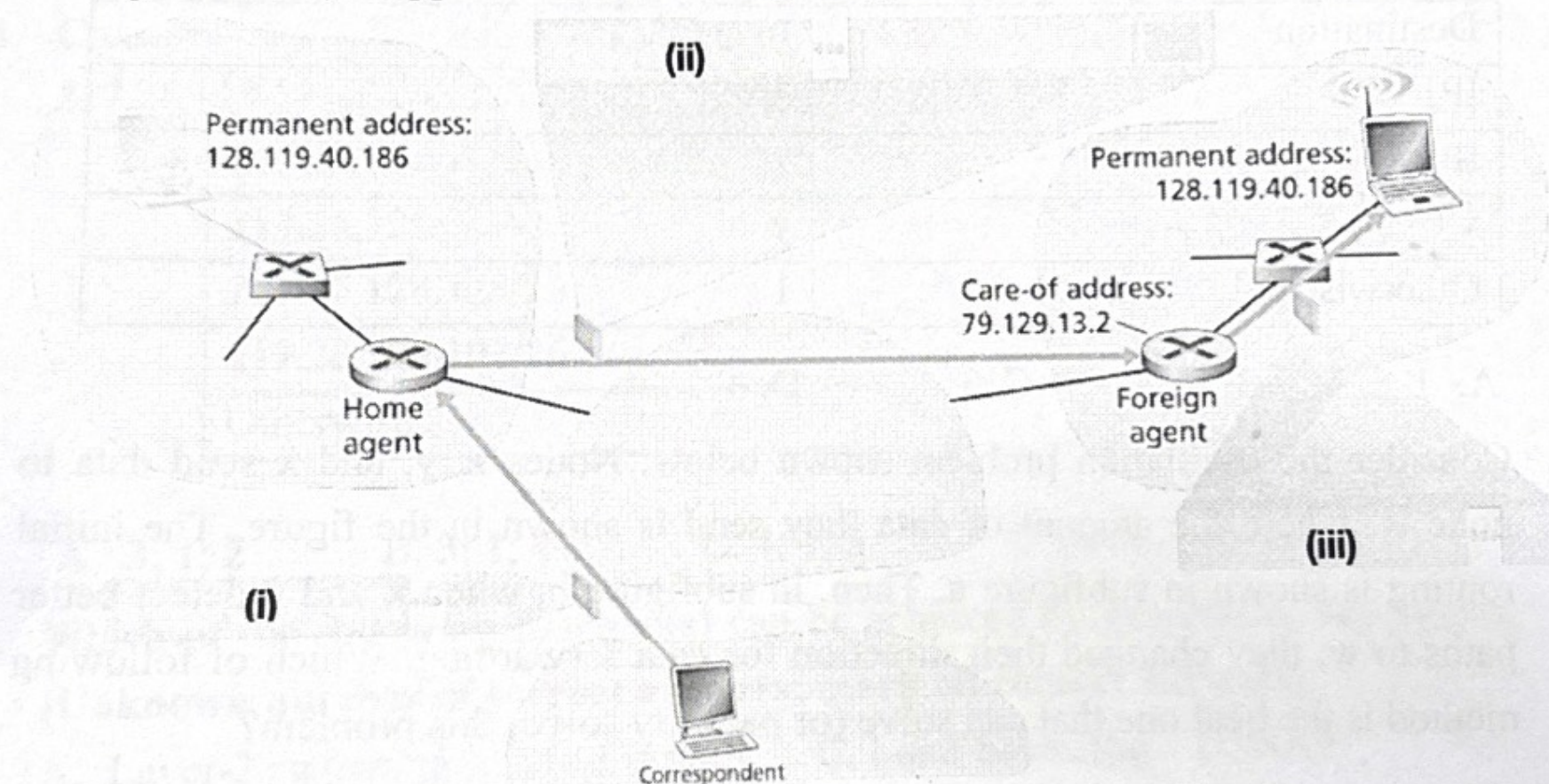


- A. Let nodes update their paths in an asynchronous fashion with a pre-defined order; meanwhile, every node waits for a fixed duration before the next updates.
 B. Let nodes update their paths at randomized time.
 C. Let nodes sense collisions before updating.
 D. Let nodes detect collisions while updating.

10. Which of the following device or protocol is NOT plug-and-play?
- A. Switch B. Routers C. ARP D. DHCP

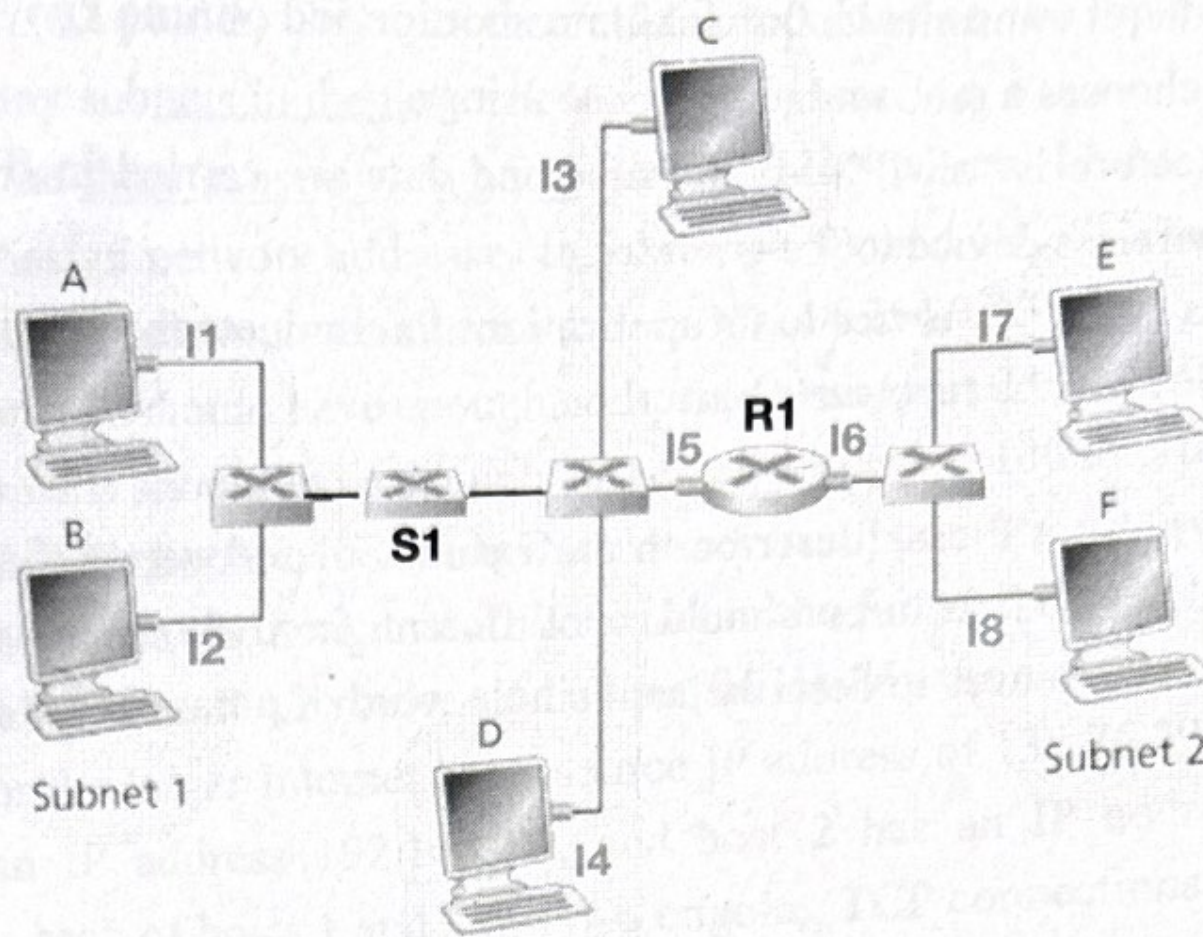
Problem III. Fill in the blanks (1 point each blank)

- Five active nodes—nodes A, B, C, D and E—compete for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability p . The probability that there is a successful transmission in a time slot is _____.
- Consider two-dimensional parity checking with even parity scheme. Each row contains 5 bits data together with 1 checking bit. Suppose a receiver receives 18 bits of data 101011101100010111. There is at most one bit error. The 18 bits of data that the sender sends is _____.
- Consider that a host has moved from home network to visited network (see the figure below). Suppose indirect routing is applied.



Fill in the blanks with the destination IP address of the associated datagrams:

- (i) _____; (ii) _____; (iii) _____
- Suppose hosts A and B has already established a TCP connection. The maximum segment size (MSS) is 1KB, and round-trip time (RTT) between A and B is 4ms. Suppose there is no congestion occurs. It is recently at congestion avoidance state. It takes _____ms for the congestion window to increase from 8KB to 32KB.
 - Consider the network topology **on the top of next page**. Notations "I1, I2, ..." denote the interfaces. Suppose host A would like to send an IP datagram to host E. Assume that every interface's ARP cache is empty (unless otherwise specified), and every switch table is empty.



- Which interface's MAC address will A perform an ARP query to find? _____
- Which interface(s) will receive the ARP query? (Please list all) _____
- Which interface(s) will send back an ARP response? (Please list all) _____
- Which interface(s) will receive the ARP response? (Please list all) _____
- After host A sent the IP datagram to host E, host B then sends an IP datagram to host D. Suppose host B already knows host D's MAC address. Which interface will receive this IP datagram? (Please list all) _____
- After host B sends an IP datagram to host D, host A then sends an IP datagram to host B. Will switch S1 receive the datagram? _____
- What is the gateway of Subnet 2? The IP address of _____

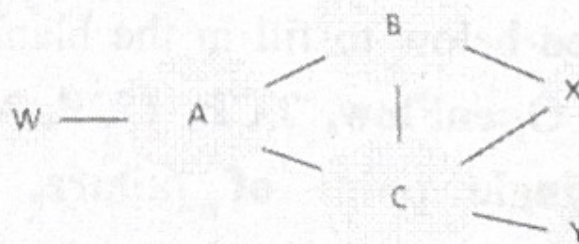
Please use the best term listed below to fill in the blanks: **OSPF, ARP, Ethernet, 802.11, 2G, 3G, 4G, BGP, OpenFlow, TCP, UDP, VLANs, RTS, CTS, DIFS, SIFS, hidden terminal, single point of failure, fading, collision, binary exponential backoff, synchronization, three-way handshaking, four-way handshaking, reliable data transfer, congestion control, flow control, tunneling, transport, link, network, physical, application.** Each term is used at most once.

- The Internet's _____ layer is responsible for moving packets known as datagrams from one host to another. The _____ layer is responsible for moving a packet from one node (host or router) to the next node in the route.
- 802.11 does not implement CSMA/CD due to _____ and _____ issues.
- _____ has been most widely adopted to achieve IPv4-to-IPv6 transition.
- A switch that supports _____ allows multiple virtual local area networks to be defined over a single physical local area network infrastructure.
- Ethernet frame begins with an 8-byte preamble field, containing 7 bytes of 10101010 and one byte of 10101011. This field is used for _____.
- IEEE 802.11 protocol allows a wireless host to use a short _____ control frame to reserve access to the channel. Meanwhile, if initially the wireless host senses

- the channel idle, it transmits its frame after a short period of time known as ____; otherwise, it chooses a random backoff time using ____.
12. ____ architecture is “all-IP”—both voice and data are carried in IP datagrams to/from the wireless device to the gateway.
13. TCP provides ____ service to its applications to eliminate the possibility of the sender overflowing the receiver’s buffer.

Problem IV. (6 points) Please describe three **features** of software defined network (SDN). Those described features should be different from those in conventional routing. Please use sentences to describe, and single word or phrase is not acceptable.

Problem V (5 points) Consider the following figure. W, X, and Y are access ISPs, and A, B, and C are backbone ISPs. You can regard W, X, Y, A, B, C as autonomous systems (ASs). Suppose there is a content provider (i.e., the company which provides contents such as videos and files) at W, which forward contents to hosts at X and Y.



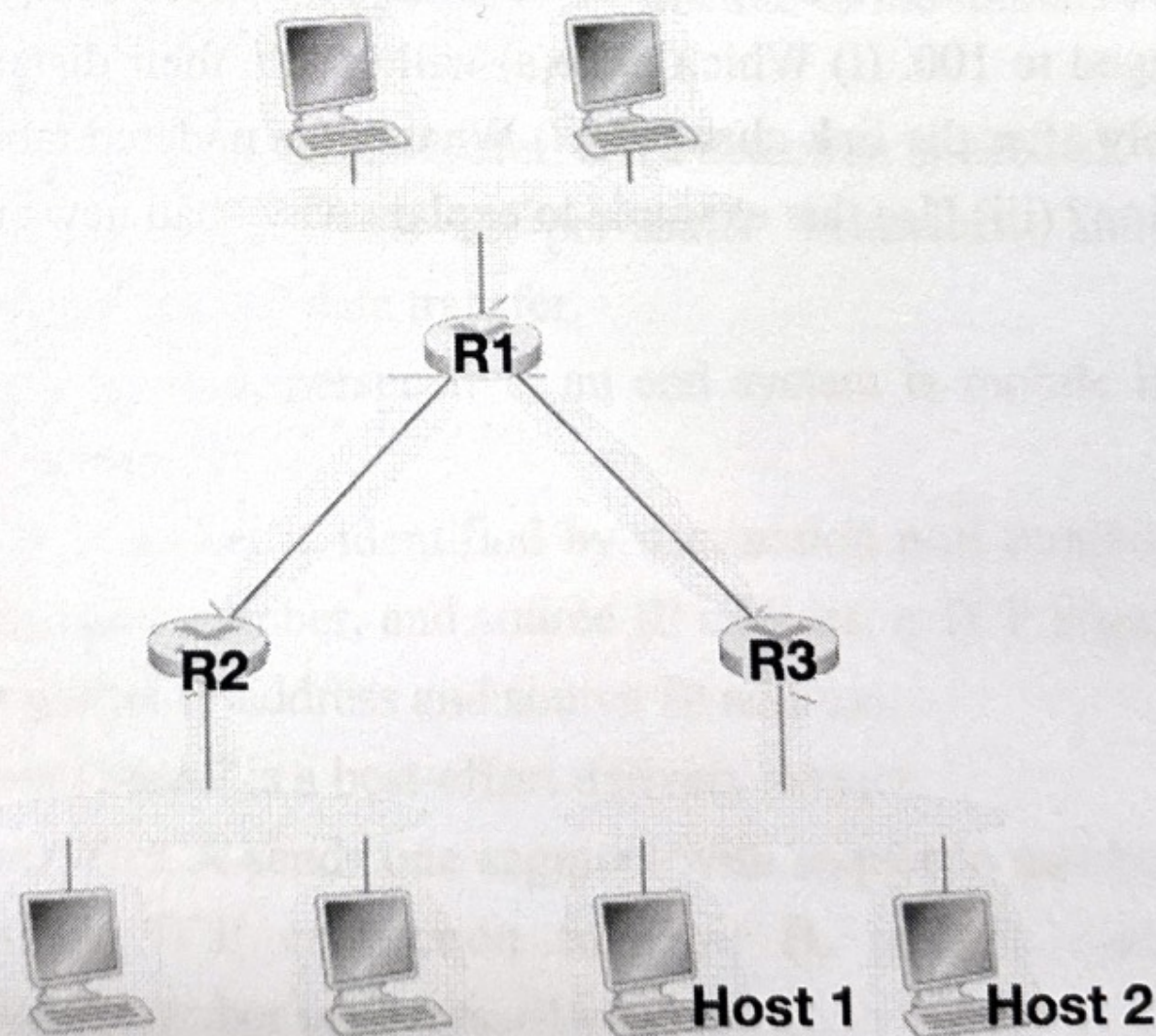
- a) If X does not want to help B to forward the traffic destined to Y, then how would the routers in X do? (Hint: BGP routing)
- b) Even under the scenario in a), there are some applications for which the contents provided by W may go to X first then flow to Y. Identify one such application.

Problem VI. (12 points) Consider the network topology on the top of next page.

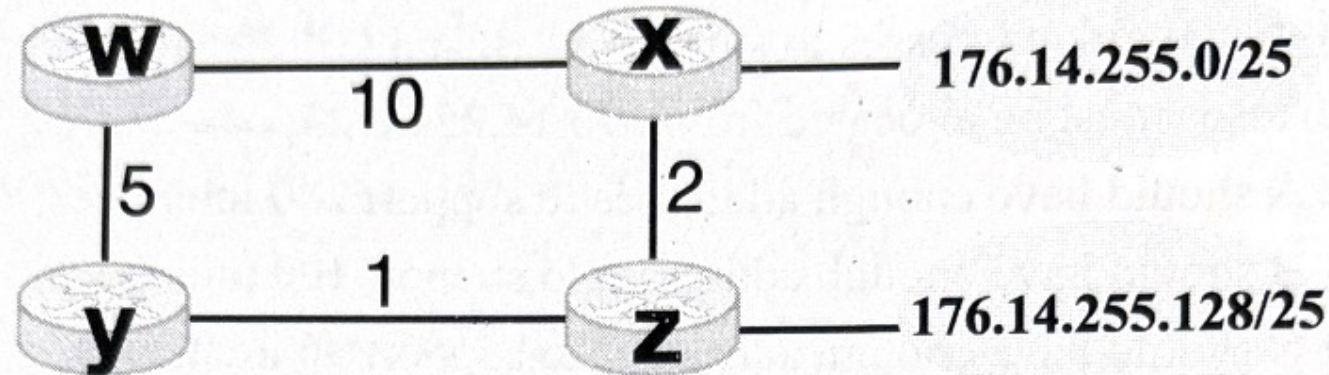
- a) How many subnets in the network topology in total?
- b) Denote the three subsets with hosts (starting clockwise at 12:00) as subnets A, B, and C. Assign network addresses to subnets A, B, and C:
- All addresses must be allocated from 176.14.254.0/23;
 - Subnet A should have enough addresses to support 240 interfaces;
 - Subnet B should have enough addresses to support 100 interfaces;
 - Subnet C should have enough addresses to support 96 interfaces.

For each subnet, the assignment should be in the form of a.b.c.d/x.

- c) Suppose R3 is an NAT router, and all traffic leaving the subnet containing hosts 1 and 2 for the larger Internet has a source IP address of 138.76.29.7. Suppose host 1 has an IP address 192.168.1.2, and host 2 has an IP address 192.168.1.3. Suppose each of hosts 1 and 2 has two ongoing TCP connections, all to port 80 at host 128.119.40.86. Provide the four corresponding entries in the NAT translation table. Note: For the information (e.g., port number, IP address) that was not given, you can pick any arbitrary value as long as it is feasible/reasonable.



Problem VII (12 points). Consider a network as shown in the figure below. Distance vector algorithm is used to calculate the forwarding table.



- Draw the distance vector table of node y after convergence **WITHOUT** poisoned reverse. You do not need to draw those tables before convergence.
- Draw the distance vector table of node y after convergence **WITH** poisoned reverse. You do not need to draw those tables before convergence.
- Based on b), draw the forwarding table of node y that contains two entries related to network prefix 176.14.255.0/25 and 176.14.255.128/25. Note: To denote an interface, you can use notation (a,b) to indicate the interface of a connected to b.
- Consider the case **WITHOUT** poisoned reverse. Suppose $\text{cost}(y,z) = \text{cost}(z,y)$ has been changed to 100. (i) Which node(s) will update their distance vector table(s) immediately after the link change? (ii) What is the updated table of node y in the first iteration? (iii) Use this example to explain why “bad news travels slowly”.